DETSI Report

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Background

Accurately characterizing drivers and resilience of seagrass cover is essential to improving the ecological health of the Moreton Bay and Wide Bay ecosystems. Over the past few decades, various scientific efforts have informed our current understanding of trends, behaviors, and flood responses of seagrass cover in both Moreton Bay and Wide Bay [Preen et al., 1995, McMahon et al., 2005, York et al., 2022] [Abal and Dennison, 1996, Udy and Dennison, 1997, Samper-Villarreal et al., 2016, Ovsyanikova et al., 2026] as well as Australia more widely

[Kilminster et al., 2015, Gilby et al., 2018, O'Brien et al., 2018]. Remote sensing has offered new capabilities for seagrass cover mapping [Roelfsema and Phinn, 2009], while innovative statistical approaches have offered novel methods for studying resilience at site-specific scales [Gilby et al., 2023]. Our work contributes to this body of literature by applying spatially explicit statistical models [Cressie, 1993] to study seagrass cover drivers, resilience, and recovery in both Moreton Bay and Wide Bay. Specifically, we provide and contextualize answers to four specific research questions, applicable to both Moreton Bay and Wide Bay:

1. Can we identify water quality covariates that act as triggers for seagrass cover management responses? Are these triggers different based on whether they result from a specific event (e.g., flood) or are ambient?
2. Did seagrass cover improve after the flood event(s)? If so, how and where?
3. Can we identify site-specific resilience indicators, which describe where recovery is likely to be enhanced or impeded?
4. What factors contribute to seagrass resilience?

The spatially explicit statistical models (i.e., spatial models) we used to answer these research questions are related to commonly used linear and generalized linear models but incorporate *spatial dependence* Spatial dependence measures how similar two sites are based on their distance. Formally, we can build spatial dependence directly into spatial models by leveraging Tobler's First Law of Geography [Tobler, 1970], which states succinctly that "nearby sites tend to be more similar than distant sites". Not only is this intuitive based on "first principles" perspective, but spatial models to provide many advantages for modeling spatial data compared to models that assume independence among observations (i.e., nonspatial models). First, nonspatial models tend to underestimate uncertainty, yielding covariate standard errors and p-values for covariates that are too small and corresponding confidence intervals that are too narrow. Second, spatial models improve prediction at unsampled locations by borrowing strength from nearby observations, yielding more precise predictions and corresponding standard errors that vary by location. Third, spatial models decompose model error into separate spatial and nonspatial components, informing residual and resilience structure in ways that nonspatial models cannot. Fourth, spatial models can provide ecologically meaningful insight into the spatial dependence structure of the process, informing relationships among measured and unmeasured variables as well as future sampling plans. Zimmerman and Ver Hoef [2024] provide a thorough review of spatial models for ecological and environmental data, providing more detail and justification for the claims made above.

Spatial models offer many benefits over commonly used machine learning algorithms like random forests [Breiman, 2001, Cutler et al., 2007, James et al., 2017, Kuhn and Silge, 2022]. First, traditional machine learning algorithms do not incorporate spatial dependence and, by consequence, often perform worse than spatial models at characterizing important covariates and predicting at unobserved locations [Fox et al., 2020]. Second, machine learning algorithms struggle to quantify uncertainty in a principled manner, something spatial models excel at. Third, covariate inference is more straightforward using spatial models, as they naturally provide p-values that enable structured testing of hypotheses. Fourth, spatial models provide a

rich set of diagnostics like leverage and Cook's distances [Montgomery et al., 2021], quantities not generally defined for machine learning algorithms. This is not to say that machine learning algorithms cannot be very useful in ecological data analyses, but rather than spatial models are often a more effective tool for modeling spatial data.

Though there are many benefits of spatial models over machine learning algorithms, the reverse is also true. Machine learning algorithms much more simply capture complex nonlinearities "out-of-the-box", bypassing the tuning often required for spatial models that involve nonlinear components like splines. This can be especially useful for ecological data, which are often full of nonlinear relationships [Fox et al., 2017]. Machine learning algorithms also typically rely on few, if any, distributional assumptions and can be more robust against unusual observations like outliers [Hastie et al., 2009]. Machine learning algorithms also tend to be much mroe computational efficient than spatial models. Importantly, machine learning algorithms and spatial models can be complementary tools that help shed insight into complicated ecological processes (something we reiterate in the upcoming analyses). An exciting and developing area of statistical research involves incorporating spatial modeling strategies directly into machine learning algorithms [Saha et al., 2023, Heaton et al., 2025].

The first spatially explicit model we define is the spatial linear model. The spatial linear model is a *mixed-effects model* [Pinheiro and Bates, 2000, Bolker, 2015] written as

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + ... + \beta_p x_{p,i} + \tau_i + \epsilon_i, \tag{1}$$

where $i = 1, 2, ..., n$ indexes the n observations. The quantity y_i is the response variable (for observation i), the β parameters are slope parameters controlling the covariates, $x_{j,i}$ (for $j = 1, 2, ..., p$). The random error τ_i is a spatial random error, and the random error ϵ_i is the nonspatial (i.e., independent) random error. The spatial random error is spatially structured, capturing spatial residual effects like environmental gradients we do not have data to represent. The nonspatial random error is not spatially structure and captures nonspatial residual effects like measurement error and microscale variation. The difference between a traditional (i.e., nonspatial) linear model and a spatial linear model is the inclusion of τ_i (in the spatial linear model). The spatial random error implied by τ_i is governed by a *spatial covariance function*. The spatial covariance function allows the model to incorporate spatial dependence, yielding the improved model performance previously described. Figure 1 shows three different spatial covariance functions that describe different ways the spatial covariance decays with distance. Model parameters and standard errors are estimated from data using likelihood-based [Patterson and Thompson, 1971, Harville, 1977, Wolfinger et al., 1994] or semivariogram-based [Cressie, 1985, Curriero and Lele, 1999] estimation methods. Often, the spatial linear model in Equation 1 in is written more compactly using *matrix notation*:

$$\mathbf{y} = \mathbf{X}\beta + \tau + \epsilon, \tag{2}$$

where $\mathbf{X}$ is a matrix that holds the j covariates for each of n observations, and $\mathbf{y}$, β, τ, and τ are vectors that contain each of the n or j quantities from Equation 1. The spatial linear

model can be adjusted to account for nonspatial random effects just as one would add random effects to a nonspatial linear mixed model:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{u} + \tau + \epsilon, \tag{3}$$

where $\mathbf{Z}$ is a random effect design matrix (dimension $n \times k$) that indexes each of the k random effects in $\mathbf{u}$. For more information about specific forms of various spatial covariance functions, see Schabenberger and Gotway [2017] and Zimmerman and Ver Hoef [2024].

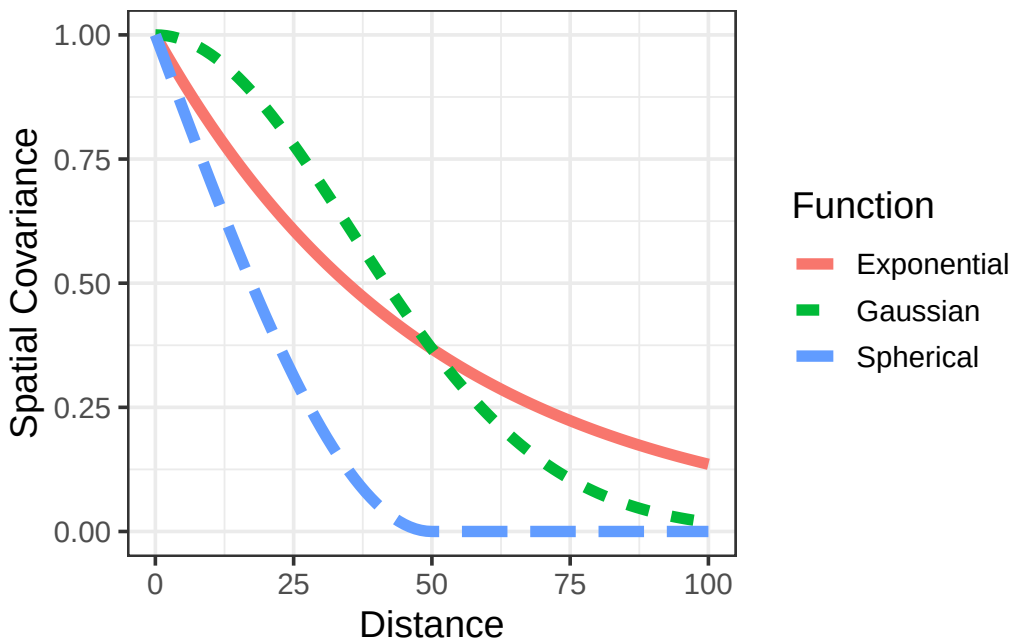

Figure 1: Three spatial covariance functions that represent different spatial dependence behavior.

Like the traditional linear model, the spatial linear model can capture complex nonlinearities in the data through the use of interaction effects [Faraway, 2002, Lenth and Piaskowski, 2025], splines [Chambers and Hastie, 1992], additive models [Wood, 2017, Pedersen et al., 2019], and penalized regression models like the ridge and lasso [Tibshirani, 1996]. Generally, this is accomplished by leveraging *basis functions* that break down nonlinear behavior into a set of additive linear components that may include penalty terms which guard against overfitting. Furthermore, there is no assumption that y_i itself is normally distributed; rather, distributional assumptions are placed on the random errors after incorporating the slope parameters. This clarification implies the spatial (and nonspatial) linear models can be quite effective at modeling a range of data that are not normally distributed. For highly nonnormal data, tools like generalized linear models, which we discuss later, are helpful. Finally, under general conditions, the slope parameters of the spatial (and nonspatial) linear models are approximately

normally distributed with a large sample size (see Casella and Berger [2024] for a general justification via the Central Limit Theorem and Zhang and Zimmerman [2005] for nuances regarding spatial data). This is beneficial because it justifies valid hypothesis testing on the slope coefficients (often of primary ecological interest), no matter the distribution of the data (given certain assumptions are satisfied).

One of the most useful applications of the spatial linear model is predicting the response variable at unobserved locations. A very useful spatial prediction method called Kriging has a rich theoretical justification [Cressie, 1990, Stein, 1999]. Kriging is the spatial analogue to best linear unbiased prediction (BLUP) for mixed models [Henderson, 1975]. Via Kriging, the spatial linear model produces BLUPs at each unobserved location of interest, alongside prediction intervals to characterize uncertainty. While useful, sometimes the goal is to create a BLUP for an entire region of interest. This technique, which builds upon the theoretical foundations of Kriging, is called block Kriging [Ver Hoef, 2002, Cressie, 2006]. The intuition is to make point predictions at a very dense grid in the study region and then average these predictions, appropriately characterizing uncertainty. Together, Kriging and block Kriging provide a complete set of tools for spatial prediction, given the fit of a spatial linear model.

Spatial generalized linear models are an extension of spatial linear models to highly nonnormal data [Christensen and Waagepetersen, 2002, Zhang, 2002, Diggle and Ribeiro Jr, 2007, Bonat and Ribeiro Jr, 2016]. Ver Hoef et al. [2024] proposed a novel application of the Laplace approximation to enable restricted maximum likelihood estimation of spatial generalized linear models. Their model formulation builds upon Equation 1 and is given by

$$g(\mu_i) = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + ... + \beta_p x_{p,i} + \tau_i + \epsilon_i, \tag{4}$$

where $g(\mu_i)$ is a "link function" that links the distribution of y_i to a linear function of its mean, μ_i. Common link functions include the log link for count and skewed data and the logit link for binomial and proportion data (Table 1). Using the same argument as for spatial linear models, nonspatial random effects can be added to spatial generalized linear models.

Table 1: Common generalized linear model families, link functions, and appropriate data types.

Family	Link Function	Link Name	Data Type
Binomial	$f(\mu) = \log(\mu/(1-\mu))$	Logit	Binary; Binary Count
Poisson	$f(\mu) = \log(\mu)$	Log	Count
Negative Binomial	$f(\mu) = \log(\mu)$	Log	Count
Beta	$f(\mu) = \log(\mu/(1-\mu))$	Logit	Proportion
Gamma	$f(\mu) = \log(\mu)$	Log	Skewed
Inverse Gaussian	$f(\mu) = \log(\mu)$	Log	Skewed

Spatial models are complicated, both theoretically and computationally. This has contributed to their relative lack of use in the ecological community. However, recent advances in open-

source software have made spatial models more accessible to ecologists who have strong subject matter expertise but may lack sufficient statistical background or computational skills to implement them from scratch. One such advancement is the `spmodel` **R** package [Dumelle et al., 2023], which emulates base-**R** functions like `lm()` and `glm()` to account for spatial dependence via `splm()` and `spglm()`, respectively. Familiar **R** functions like `summary()` and `predict()` acn be used directly on models fit via `spmodel`. Importantly, `spmodel` also has tools for spatial random forest residual modeling [Fox et al., 2020], applications to large data of several thousand observations via spatial indexing [Ver Hoef et al., 2023], and cross validation [Stone, 1974, Roberts et al., 2017].

Moreton Bay

The Data

We started with just over 10,000 seagrass cover observations in Moreton Bay across three financial years (2020-2021, 2022-2023, and 2023-2024). eReefs data [Steven et al., 2019] with modeled water quality covariates related to turbidity and salinity were only available at 7,657 observaations. Because one of our research questions involved connecting water quality covariates to management actions, we proceed with this subset of 7,657 observations for modeling. Of the 7,657 observations, 1,995 were in 2020-2021 (pre-flood), 3,329 were in 2022-2023, and 2,333 were in 2023-2024 (Table 2).

Table 2

Financial Year	Sample Size	Percent (of Total)
2020-2021	1,995	26.1%
2022-2023	3,329	43.5%
2023-2024	2,333	30.5%
Total	7,657	100.0%

Total seagrass observations Moreton Bay by financial year.

Water Quality and Geographic Modeling

Total cover is a percentage that ranges from 0 (no seagrass) to 100 (seagrass entirely contained in an area). Figure 2 shows total cover by financial year throughout the Bay. Total cover tends to be largest along the eastern shoreline of the Bay and lowest in the middle of the Bay. Figure 2 also reveals that 2022-2023 had by far the most sampling in the middle of the Bay.

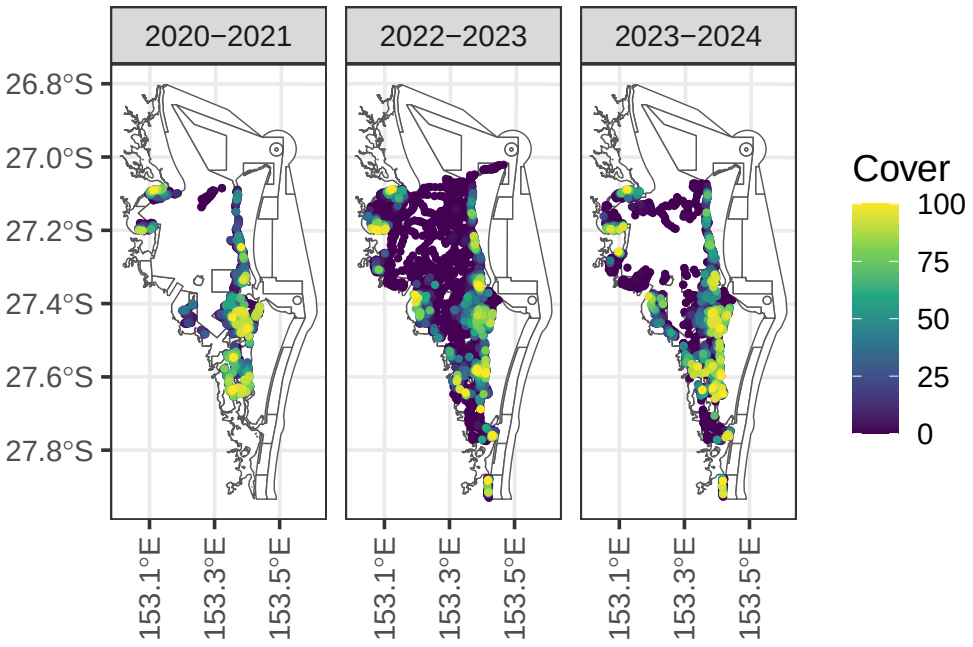

Figure 2: Total seagrass cover in Moreton Bay by financial year.

We considered for modeling several water quality and geographic covariates. Table 3 summarize the water quality variables, which included both ambient and flood-related metrics related to turbidity, salinity, and temperature. The "Variable" column indicates the water quality variable turbidity, salinity, or temperature. The "Flood" variable indicates whether the measurement corresponds to a flood variable or not. The "Period" variable indicates the prior period (in days) relative to sampling or indicates whether the variable corresponds to a flood event. The "Type" column is the type of statistic, either a mean, minimum, maximum, number of days above (or below) a cutoff during "Period", or a maximum spell (i.e., duration) above (or below) a cutoff during "Period". The "Included" variable indicates whether the variable was included as a covariate in the final spatial model. We included only a few variables in the final model: mean turbidity in the 90 days prior to sampling, mean turbidity during the flood event, and mean temperature in the 90 days prior to sampling. We only included a single turbidity metric prior to sampling because these variables were all quite correlated with one another, making it hard for the model to tease apart signals. We believed 90 day mean turbidity prior to sampling to be the most accurate "running average" of turbidity prior to sampling. We used a similar rationale to select only a single turbidity metric from the flood and a single temperature metric prior to sampling. We omitted a flood salinity metric because it did not appear to have a significant impact on the total cover at modeling, and additionally, because we lacked data for salinity prior to sampling, it would be challenging to establish a "baseline" for salinity in order to tease apart prior to sampling effects from flood effects. We provide more detail regarding the modeling in Section .

Table 3: Moreton Bay water quality variables considered for modeling.

Variable	Flood	Period	Type	Cutoff	Included
Turbidity	No	7	Mean	NA	No
Turbidity	No	30	Mean	NA	No
Turbidity	No	90	Mean	NA	Yes
Turbidity	No	7	Above	3	No
Turbidity	No	30	Above	3	No
Turbidity	No	90	Above	3	No
Turbidity	No	7	Above	5	No
Turbidity	No	30	Above	5	No
Turbidity	No	90	Above	5	No
Turbidity	No	7	Above	6	No
Turbidity	No	30	Above	6	No
Turbidity	No	90	Above	6	No
Turbidity	Yes	Flood	Above	3	No
Turbidity	Yes	Flood	Above	5	No
Turbidity	Yes	Flood	Above	6	No
Turbidity	Yes	Flood	Duration	3	No
Turbidity	Yes	Flood	Duration	5	No
Turbidity	Yes	Flood	Duration	6	No
Turbidity	Yes	Flood	Mean	NA	Yes
Turbidity	Yes	Flood	Max	NA	No
Salinity	No	7	Below	20	No
Salinity	No	30	Below	20	No
Salinity	No	90	Below	20	No
Salinity	No	7	Below	25	No
Salinity	No	30	Below	25	No
Salinity	No	90	Below	25	No
Salinity	No	7	Below	30	No
Salinity	No	30	Below	30	No
Salinity	No	90	Below	30	No
Salinity	Yes	Flood	Below	20	No
Salinity	Yes	Flood	Below	25	No
Salinity	Yes	Flood	Below	30	No
Salinity	Yes	Flood	Duration	20	No
Salinity	Yes	Flood	Duration	25	No
Salinity	Yes	Flood	Duration	30	No
Salinity	Yes	Flood	Mean	NA	No
Salinity	Yes	Flood	Min	NA	No
Temperature	No	7	Mean	NA	No
Temperature	No	30	Mean	NA	No

Table 3: Moreton Bay water quality variables considered for modeling.

Variable	Flood	Period	Type	Cutoff	Included
Temperature	No	90	Mean	NA	Yes

Table 4 summarizes the geographic variables, which have the following structure:

- The "Sediment Type" variable has six levels: Sand, Sandy Mud, Muddy Sand, Mud, Rock, and Other.
- The "Tidal Range" variable has three levels: Shallow Subtidal, Intertidal, and Deep Subtidal.
- The "Years Since Flood" is numeric from zero (pre flood) to 1.93 years post flood (February, 2024).
- The "Nearest River" variable has six levels: Brisbane River, Caboolture River, Logan River, Ningi Creek, Pimpama Coomera and Nerang Rivers, and Pine River.
- The "Distance to (Nearest River) Outlet (in km)" variable is numeric from 0.24 to 34.97.
- The "Growing Season" variable has two levels: Growing (September - December) and Not Growing (January - August).
- The EHMP subzone variable has eleven levels: Bramble Bay, Broadwater, Central Bay, Deception Bay, Eastern Banks, Moreton Bay Open Coastal Waters, North-Eastern Moreton Bay, Northern Banks Open Coastal Waters, South-Eastern Moreton Bay, Southern Bay, and Waterloo Bay (Figure 3).

Table 4

Variable	Effect	Included
Sediment Type	Fixed	Yes
Tidal Range	Fixed	Yes
Years Since Flood	Fixed	Yes
Nearest River	Fixed	No
Distance to Outlet	Fixed	No
Growing Season	Fixed	No
EHMP Subzone	Random	Yes

Moreton Bay geographic variables considered for modeling.

From Table 4, we kept the sediment type, tidal range, and years since flood variables as covariates in the model. The nearest river and distance to nearest outlet variables were removed, as they did not seem related to total cover. We removed growing season from the model after determining that it likely it not ecologically useful, as a simple month proxy likely does not adequately account for the complexities of the more nuanced aspects of seagrass growth with

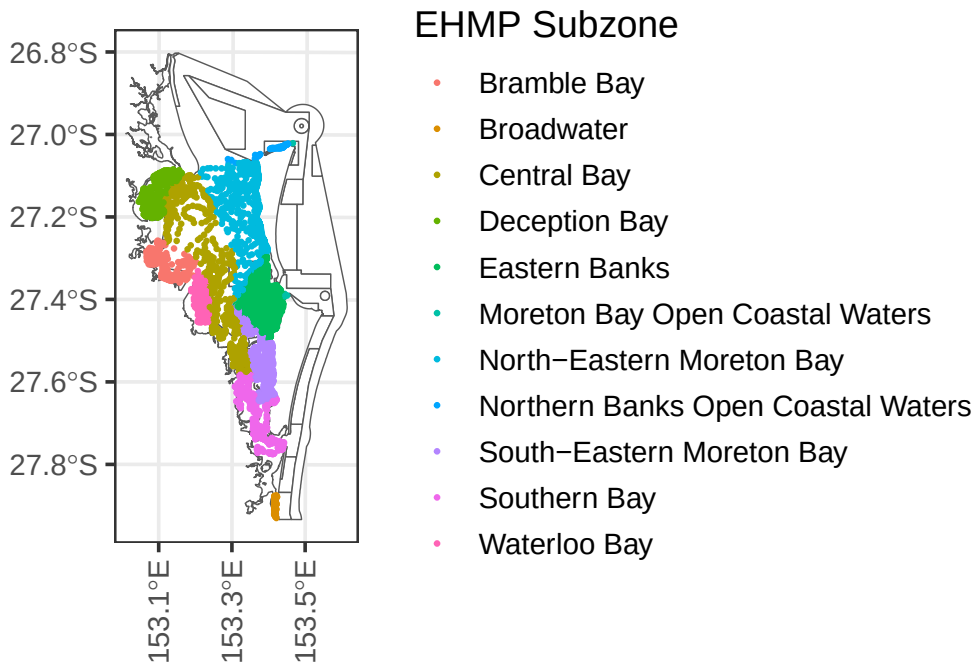

Figure 3: EHMP Subzones in Moreton Bay.

time. We used EHMP subzone as a random effect to account for within-subzone correlation that may impact the relationships among total cover and the remaining variables. We provide more detail regarding the modeling in Section .

Model Fit

Table 5

Variable	Type	No. Levels
90 Day Temperature	Numeric	NA
90 Day Turbidity	Numeric	NA
Mean Flood Turbidity/Years Since Flood^2	Numeric	NA
Sediment Type	Categorical	6
Tidal Range	Categorical	3

Variables in the final Moreton Bay total cover model, their type (numeric or categorical), and if categorical, the number of unique levels.

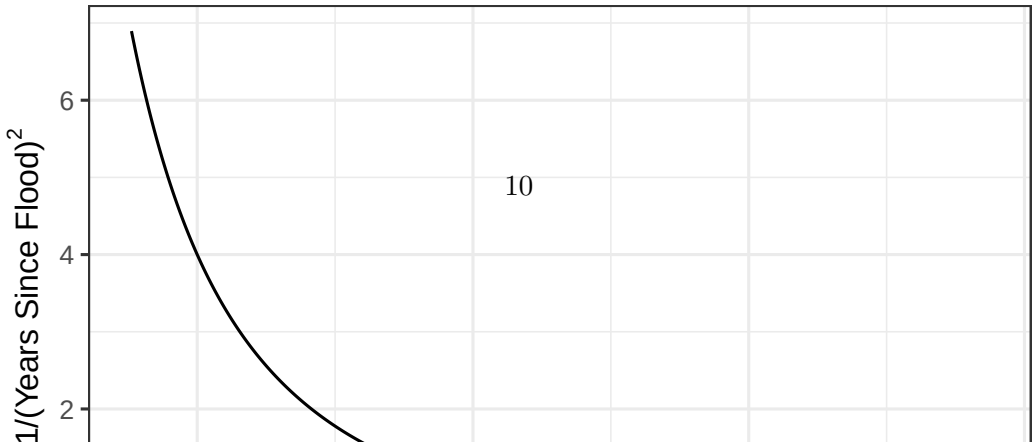

Table 6: ANOVA table of covariates in the Moreton Bay water quality and geographic model.

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Error in `mutate()`:
i In argument: `effects = c(...)`.
Caused by error:
! `effects` must be size 7 or 1, not 5.
```

temperature (prior to sampling), pre- vs post-flood, years since the flood, 90 day turbidity (prior to sampling), and mean turbidity during the flood period scaled by time since the flood squared (Table 5). We scaled mean turbidity during the flood period so that the effect of turbidity during the flood event was allowed to decay with time since the flood (Figure 4). Intuitively, this means that with time since the flood, the impact of turbidity during the flood event is lessened. Additionally, we used an exponential spatial covariance function and included a random effect for EHMP subzone. We fit the model using restricted maximum likelihood via the `splm()` function in `spmodel`.

Table 6 shows the results from an analysis of variance applied to the fitted model. All explanatory variables had large test statistics and small p-values (p-value < 0.01); here are some relevant takeaways:

- Tidal Range: The estimated (marginal) average total cover is largest in the intertidal (26.89%), followed by shallow subtidal (16.51%) and deep subtidal (4.88%).
- Sediment Type: The estimated (marginal) average total cover is largest in the other (25.6%), followed by muddy sand (19.1%), sandy mud (17.8%), mud (14.3%), sand (14.2%), and rock (5.62%).
- 90 Day Temperature: A one-degree (C) increase in 90 day temperature is associated with a decrease in average total cover by 0.88%.
- Mean Flood Turbidity / (Years Since Flood)^2: An one-unit increase in mean turbidity during the flood is associated with a decrease in average total cover by 2.05%/(Years Since Flood)^2.
- 90 Day Turbidity: A one-unit increase in 90 day turbidity is associated with a decrease in average total cover by 7.18%.

The estimated marginal means for tidal range and sediment type were calculated using the `emmeans` **R** package [Lenth and Piaskowski, 2025].

We may inspect model diagnostics to better understand the overall model fit. Here we discuss how to partition the model into different components of variability, infer decay behavior of spatial autocorrlation, analyze standardized residuals to assess model assumptions, and identify influential observations that have a sizable impact on model fit. From Table 7, we see that of the total model variability, 11.2% is attributable to the explanatory variables (this quantity is sometimes called the Pseudo R-squared), 31.6% is attributable to the spatial random error,

7.3% is attributable to the EHMP subzone, and the remaining variability is independent random error. The estimated exponential spatial range (ϕ) is 745.5 meters, which suggests two total cover observations are approximately (spatially) uncorrelated at a distance of $3\phi =\approx$ 2km, or two kilometers. While the raw residuals will inform our resilience modeling in Section , we can check certain model assumptions using the standardized residuals, which have been decorrelated. Generally, the standardized residuals are centered around zero, though there is a slight right skew suggesting some observed total cover values are larger than expected (according to the model). Observations are called "influential" if they highly influence model fit. Influence is often measured via Cook's distance, and no observations exceed the often-used Cook's distance threshold value of one [Cook and Weisberg, 1982].

Table 7: Variance components in the Moreton Bay water quality and geographic model.

Variance Component	Percent (of Total Variability)	Parameter Value
Covariates	11.6%	NA
Spatial Effect	31.5%	181.4
Independent Error	49.7%	286.4
EHMP Subzone	7.3%	745.5

Block Kriging

A tremendous benefit of using the spatial linear model is that block Kriging can be directly applied to predict the mean total seagrass cover bay-wide for any combination of explanatory variables and geographic regions, even if these variables were not directly used in modeling. Figure 5 and Figure 6 show that there are several EHMP subzones and conseravtion zones that have recovered to pre-flood levels (e.g., Deception Bay, Marine National Park Zone), while others have not (e.g., Southern Bay, General Use Zone). While EHMP subzone was included in the model as a random effect, management zone type was not – via block Kriging, we can still quantify broad trends in total seagrass cover. **?** shows that total seagrass cover is recovering less effectively in the intertidal range than the shallow or deep subtidal ranges, while **?** shows that total seagrass cover is recovering bay-wide since the flood. Another useful application of block Kriging is scenario modeling, which can be used to characterize the impact of various future changes to broad-scale climatic patterns.

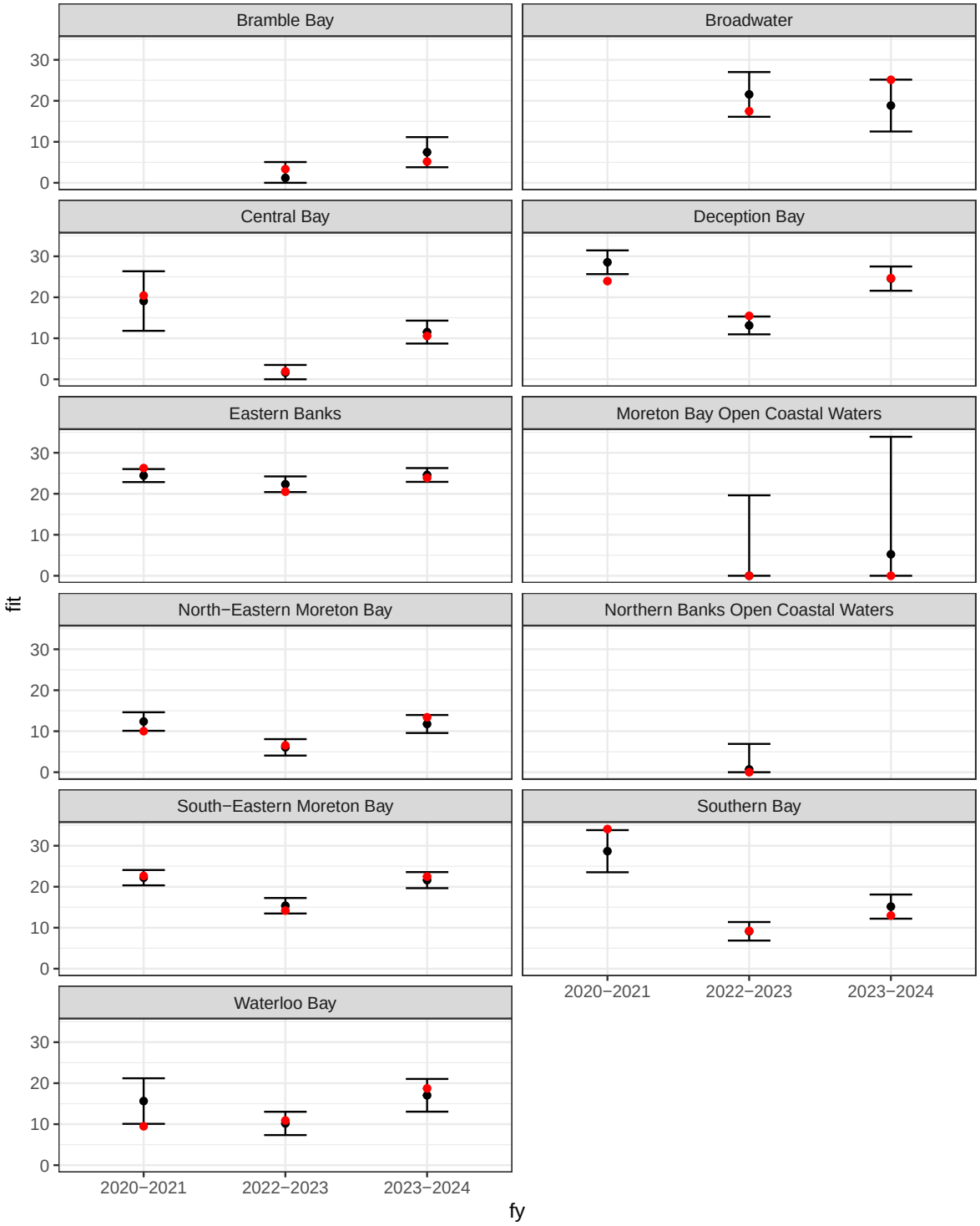

Figure 5: Average total cover predictions by EHMP subzone. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

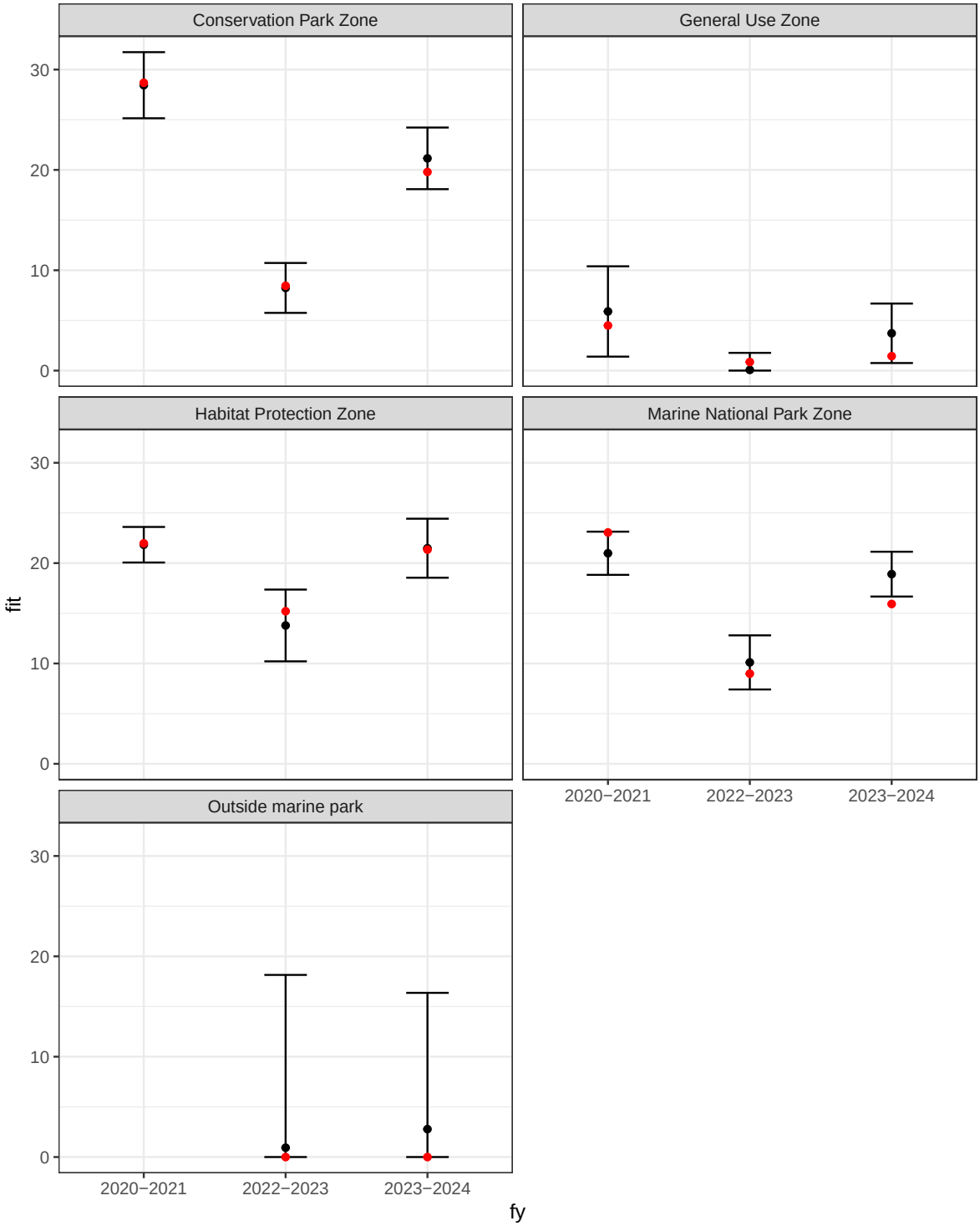

Figure 6: Average total cover predictions by EHMP conservation zone. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

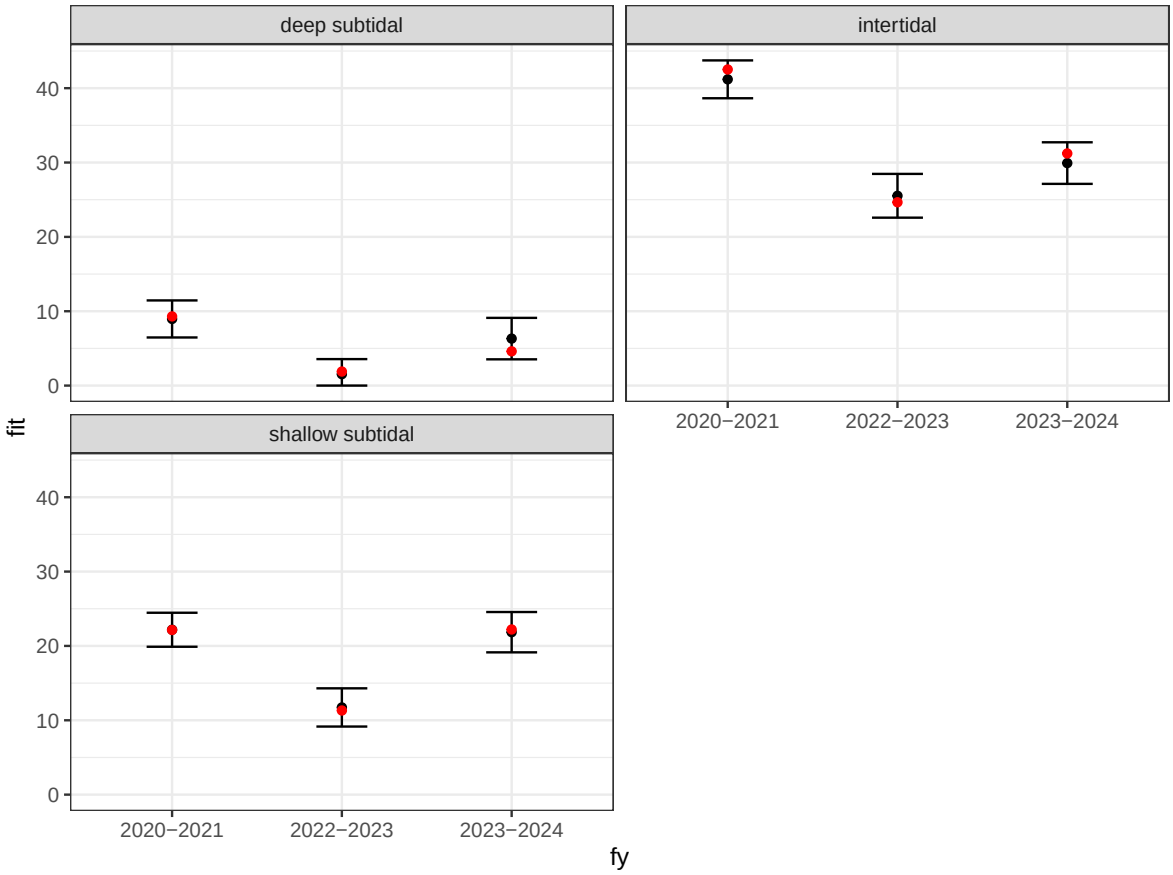

Figure 7: Average total cover predictions by EHMP tidal range. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

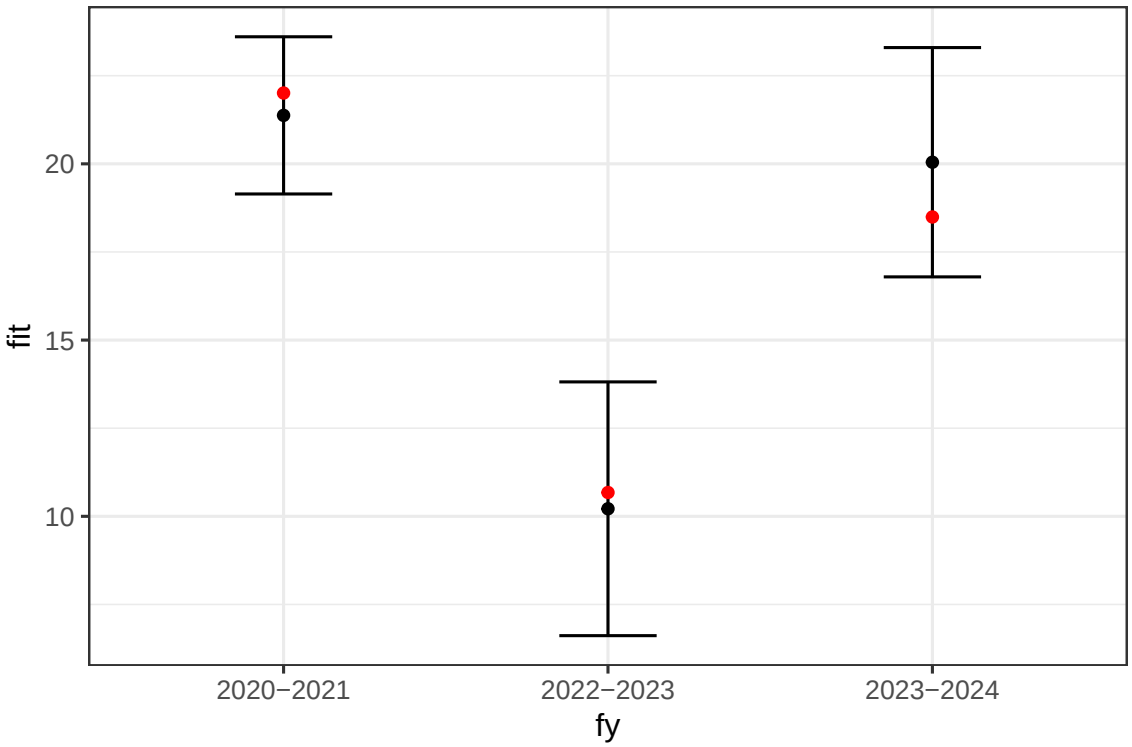

Figure 8: Average total cover predictions by Moreton Bay. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

Alternative Approaches

Resilience Modeling

Model Fit

Alternative Approaches

Wide Bay

Outstanding Questions

- What are the units for the turbidity/salinity/temperature variables?

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Appendix

Moreton Bay

Wide Bay